ZoneCast

Deployment guide for a blank machine

Ubuntu Server 24.04 LTS / 26.04 LTS

Technical manual — installation, architecture, configuration, migration, backup, upgrades and security

Application version 1.6.4 — November 2026

Table of contents

1. Introduction and what the package includes
2. Architecture and components used
3. Machine requirements
4. Preparing the blank machine
5. Method A — Native installation (systemd, recommended)
6. Method B — Docker installation
7. Application configuration (.env and the data folder)
8. First login and securing the account
9. Feature overview (zones, live announcements, schedules, custom dates, interface, alerts)
10. Migrating data from another installation
11. Security: 2FA, firewall, privileged helper, systemd hardening, HTTPS
12. Day-to-day operations and useful commands
13. Backup: manual export and automatic backup
14. Upgrading and rolling back
15. Common troubleshooting
16. Appendix — list of files provided

1. Introduction and what the package includes

ZoneCast is a PA/intercom management system for IP speakers (Fanvil A233 and, via the generic driver, other multicast brands such as Algo, CyberData, Grandstream, Axis, Tiptel, Akuvox): zones (a speaker can belong to several), schedules with holidays from multiple countries and custom dates, with overlapping schedules refused, multicast RTP playback with a conflict check for live announcements, announcement volume for supported brands, device configuration backups, persistent logs, two-factor authentication, host resource monitoring, light/dark/automatic theme, multilingual interface, full data export/import, and scheduled automatic backups kept on the server and optionally copied to an FTPS/FTP server or an SMB share.

This guide covers installation on a **brand-new Ubuntu Server LTS machine** ("blank"), both with Docker and natively (systemd), migrating data from an existing installation, backups, and upgrading an installation to a newer release (and rolling it back).

> **WHICH METHOD SHOULD I CHOOSE?**
>
> If the machine is **dedicated exclusively** to ZoneCast, the **native install (Method A)** is recommended: multicast RTP traffic reaches the physical network without having to work around the Docker bridge's NAT, and system time/network management works without the AppArmor limitations tied to containers. If the machine also hosts other services, or you'd rather be able to roll back quickly to a previous image, use Docker (Method B).

2. Architecture and components used

ZoneCast is a monolithic Python web application, designed to run on a single dedicated machine. The diagram below summarizes the main layers:

```
Browser (Bootstrap 5.3 + vanilla JS ES modules, all served locally)
        |  HTTP/JSON, strict Content Security Policy
        v
FastAPI (app/main.py) ---- routers (auth, speakers, zones, media,
        |                   playback, schedules, calendars, system,
        |                   auto_backup, backups, logs)
        v
services/ (scheduler, player, overlap check, device drivers,
           multicast_provisioning, auto_backup, backup_remote,
           audio_analysis, network_config, system_time, alerts...)
        |
        v
SQLAlchemy ORM  ---->  SQLite (data/zonecast.db, WAL mode)
                         ^
     app/migrate.py (Alembic) brings the schema up to date at
     every start, before anything else opens the database

APScheduler (in-process) runs the scheduled playbacks, the speaker
reachability check (every 5 minutes) and the automatic backup;
playback sends multicast RTP, drivers write the devices' paging lists
```

Main software components

Component	Role
FastAPI + Uvicorn	Asynchronous web framework and ASGI server

Component	Role
SQLAlchemy + SQLite	Application ORM and database (single file, no external DB server)
Alembic	Database schema migrations (`migrations/`, `alembic.ini`), applied automatically at every start — see section 14
Jinja2	Server-side page rendering (dashboard.html, login.html)
Bootstrap 5.3 + Bootstrap Icons	UI components and native light/dark theme support (data-bs-theme). Vendored under `app/static/vendor/`: no CDN, the dashboard works on a LAN without internet
Vanilla JS ES modules	`main.js` wires the feature modules in `features/`, helpers in `lib/`, plus `i18n.js` (7 languages) and `theme.js` — no build step, no inline scripts
APScheduler	In-process scheduling engine for planned playbacks, device checks and automatic backups
holidays (Python library)	Per-country holiday calendars (IT/US/GB/FR/DE/ES/JP/CN), per schedule
httpx	Async HTTP client for Fanvil device provisioning
cryptography, bcrypt, pyotp	Sensitive-data encryption, password hashing, TOTP for 2FA
smbprotocol	SMB2/3 client (pure Python) used to copy automatic backups to a network share; FTP/FTPS uses Python's standard ftplib
netplan (host)	Machine network configuration, managed by ZoneCast on request
chrony or systemd-timesyncd (host)	Time sync — ZoneCast auto-detects which of the two is installed (chrony on 26.04+, timesyncd on 24.04)

The **driver** pattern (app/services/drivers/) abstracts away the differences between speaker brands: each driver declares which features it supports (automatic multicast push, config backup, announcement volume) via simple boolean flags, so the interface automatically adapts to the selected device with no conditional code scattered across the application.

Dependencies are hash-locked: `requirements.in` lists the direct runtime dependencies and `requirements.txt` is generated from it with uv, pinning every transitive package together with its hash — pip refuses any downloaded package whose hash doesn't match. `requirements-dev.in` / `requirements-dev.txt` add the test and lint tools. Both installation methods install from `requirements.txt`; how to regenerate the files is described in CONTRIBUTING.md.

3. Machine requirements

Requirement	Recommended value
Operating system	Ubuntu Server 24.04 LTS or 26.04 LTS, minimal install
CPU	2 vCPU (1 is enough for small installations)
RAM	2 GB minimum, 4 GB recommended (an automatic backup is built in memory, in a separate process)
Disk	20 GB minimum (more if you plan to store a lot of audio files). Automatic backups are kept in data/auto_backups/ (3 copies by default), each roughly the size of the database plus the audio files if they are included; a run needs free space of at least twice its estimated size plus 512 MB
Network	Static IP on the same LAN as the speakers, with multicast routing enabled (no router/switch filtering IGMP)
Internet	Only needed while installing or upgrading (Ubuntu packages, Python packages from PyPI, Docker images). At run time the dashboard loads nothing from the internet
Outbound connections (optional)	Only for copying automatic backups off the machine: TCP 21 plus the server's passive port range (FTPS/FTP), or TCP 445 (SMB)
Access	A user with sudo privileges reachable via SSH

4. Preparing the blank machine

After installing Ubuntu Server (from the official installer, with OpenSSH enabled) and logging in over SSH for the first time:

```
# Update system packages (important: use dist-upgrade, not just
# upgrade, especially before any future major Ubuntu version upgrade)
sudo apt update && sudo apt dist-upgrade -y
sudo reboot   # if the kernel was updated

# Check the static IP address (usually configured during install
# via netplan, e.g. /etc/netplan/00-installer-config.yaml)
ip addr show

# Check that the date/time is correct and in sync
timedatectl
```

WARNING

If the machine doesn't have a static IP, speakers configured with the server's address will stop working the first time it reboots with a different IP. Configure the static IP in `/etc/netplan/*.yaml` before proceeding — or leave it to the installer and manage the final IP later from ZoneCast's own System panel (native install only, see sections 9 and 11).

5. Method A — Native installation (systemd, recommended)

Copy the package onto the machine (or clone the repository) and run the provided install script (deploy/install_ubuntu.sh):

```
scp zonecast_install_kit.tar.gz user@new-machine:/tmp/
ssh user@new-machine
mkdir -p /tmp/zonecast-src
tar -xzf /tmp/zonecast_install_kit.tar.gz -C /tmp/zonecast-src
cd /tmp/zonecast-src
sudo ./deploy/install_ubuntu.sh
```

In order, the script:

1. Installs the required system packages (python3, python3-venv, python3-pip, ffmpeg, rsync, git, sqlite3, curl)
2. Creates a dedicated, unprivileged system user, zonecast (home /opt/zonecast, no login shell), if it doesn't exist yet
3. Copies app/, migrations/, alembic.ini, deploy/ and requirements.txt (and sounds/, if present) to /opt/zonecast with rsync — data/, media/, backups/, the venv and .env are never touched
4. Generates a .env file with a random SECRET_KEY, only if one doesn't already exist
5. Creates a Python virtualenv (against the current system interpreter — important after a major Ubuntu upgrade, see note below), installs the hash-locked dependencies from requirements.txt and precompiles the bytecode as root
6. Sets ownership: all the code, the venv and deploy/ belong to root; /opt/zonecast itself is root:zonecast with mode 750, .env is root:zonecast with mode 640, and only data/, media/ and backups/ are owned (and writable) by zonecast
7. Installs the privileged helper as **/usr/local/sbin/zonecast-netctl** (root-owned, `install -m 750`) and its sudoers rule, which is first validated with `visudo -c` on a temporary file and only then installed as /etc/sudoers.d/zonecast-netctl (see section 11)
8. Installs, enables and (re)starts the systemd service (deploy/zonecast.service); at start the database schema is created or upgraded automatically
9. Preloads the sample sounds into Media (idempotent: re-running it never creates duplicates)

The script is also the **upgrade procedure**: re-running it from a newer release updates the installation in place, keeping the data and .env (section 14).

> **NOTE — OS upgrades**
>
> If you later upgrade Ubuntu's major version (e.g. 24.04 to 26.04), the default system python changes version: the existing venv will stop working ("ModuleNotFoundError") because it was built for the previous interpreter. Recreate it against the current python — the venv is owned by root: `sudo python3 -m venv --clear /opt/zonecast/venv` — then re-run `sudo ./deploy/install_ubuntu.sh` from the release folder, which reinstalls the dependencies, recompiles the bytecode and restarts the service.

Once done, check its status:

```
sudo systemctl status zonecast
sudo journalctl -u zonecast -f   # live log
```

6. Method B — Docker installation

```
sudo apt update
sudo apt install -y ca-certificates curl gnupg
curl -fsSL https://get.docker.com | sudo sh
sudo usermod -aG docker $USER   # then log back in over SSH

mkdir -p ~/zonecast
tar -xzf zonecast_install_kit.tar.gz -C ~/zonecast
cd ~/zonecast
cp .env.example .env
nano .env   # customize SECRET_KEY, admin credentials, etc.

# data folders writable by the container's user (uid 1000)
mkdir -p data media backups
sudo chown -R 1000:1000 data media backups

docker compose up -d --build
```

docker-compose.yml uses `network_mode: host`, essential so multicast RTP traffic actually reaches the physical LAN. Works only on a Linux host. In this mode the System panel shows the clock and whether the host keeps it in sync (read from the kernel, so right with chrony and systemd-timesyncd alike); network, NTP servers and timezone are managed on the host. The manual clock set still works through the SYS_TIME capability. If Docker finds the data, media or backups folder owned by root, ZoneCast stops at startup with the chown command to run.

The image (python:3.11-slim) installs the same hash-locked requirements.txt, copies app/, migrations/ and alembic.ini, keeps the code root-owned and runs the app as the unprivileged `appuser` (uid 1000), which can write only /app/data, /app/media and /app/backups — bind-mounted from ./data, ./media and ./backups. Schema migrations run at every container start, as on the native install.

7. Application configuration (.env and the data folder)

Variable	Description
SECRET_KEY	Session-cookie signing key — generated automatically by the native install script. If it is left at the example value, ZoneCast generates a random key itself and keeps it in data/session.key, so cookies are never signed with a publicly known key
DEFAULT_ADMIN_USERNAME / DEFAULT_ADMIN_PASSWORD	First admin's credentials, created only if the database is empty. The native installer generates a random password and prints it at the end; the example file (Docker) has admin / admin. Change the password on first login
DATABASE_URL	Path to the SQLite database (normally left as-is)
GLOBAL_ALL_CALL_ADDRESS / _PORT	Multicast address/port for the "all speakers" announcement (default 239.1.1.99:5004)
RTP_PAYLOAD_TYPE	G.711 encoding: 0 = PCMU, 8 = PCMA — depends on the devices
RTP_PACKET_MS / RTP_MULTICAST_TTL	RTP frame length (20 ms) and multicast TTL (8) — normally left as-is
TIMEZONE	Timezone for the schedules (default Europe/Rome) until one is applied from the System panel, which then sets both the host and the schedules
APP_PORT	Listening port, **8000 by default** (80/443 require CAP_NET_BIND_SERVICE, already granted in the native service and the Docker setup)
SESSION_MAX_AGE_SECONDS	How many seconds without use before a user's session expires (default 43200 = 12 hours); the dashboard's own automatic refreshes don't count as use. Not in .env.example: add it only if needed
MAX_UPLOAD_MB / MAX_DURATION_SECONDS	Limits on uploadable audio files (default 50 MB / 600 s)
BACKUP_MAX_BUNDLE_MB	Largest automatic backup allowed (default 500): a bigger one is refused, since it is built in memory in a child process

After editing .env: `sudo systemctl restart zonecast` (native) or `docker compose up -d --force-recreate` (Docker).

7.1 What lives in the data folder

Besides .env, some state is kept in files under data/ (/opt/zonecast/data on the native install, ./data with Docker). They matter when you move, back up or roll back an installation:

File	Purpose
zonecast.db (+ -wal, -shm)	The SQLite database, in WAL mode: the -wal/-shm companions belong to it
secret.key	At-rest encryption key for the device web passwords, the 2FA secrets and the automatic-backup passwords. Included in every export; if it is lost or replaced, those values can't be decrypted any more
session.key	Only when SECRET_KEY is left at its example value: the random session key generated in its place

File	Purpose
instance.id	A random id identifying this installation. Not part of exported bundles: it lets automatic backups tell their own files apart and pause a policy that arrived from another machine. Delete it on a VM cloned together with data/, or both would prune each other's backups at a shared destination
auto_backups/	Automatic backups kept on this server (mode 700)
zonecast.db.pre-upgrade-<revision>-<time>	Copy of the database taken right before a schema upgrade (newest 3 kept) — see section 14
*.pre-import-<time>	Database, key (and media/backups folders) set aside by an import from the dashboard — see section 10

8. First login and securing the account

1. Open http://<machine-ip>:8000 in a browser (or the port set in APP_PORT; with APP_PORT=80 simply http://<machine-ip>)
2. Log in with the credentials set in DEFAULT_ADMIN_USERNAME / DEFAULT_ADMIN_PASSWORD (native install: the password printed by the installer; Docker: admin / admin unless changed)
3. Change the password immediately from the Password button at the top (minimum 8 characters)
4. Consider enabling 2FA from the Security button
5. Create operational users from Users (admins only), with the appropriate role
6. Choose language and theme from the top bar — these are a per-browser personal preference, available to any user
7. If you want automatic backups, set them up in System > Automatic backup (section 13)

9. Feature overview

Two roles: **admin** (everything, including the Users, System, Log and Backups sections, the login alerts and direct device actions such as Apply now and Backup) and **operator** (playback, speakers, zones, media, schedules and custom dates).

9.1 Zones and speakers

A speaker can belong to several zones

Zone members are chosen in the Zones tab, from a searchable list of speakers; the speaker form shows its zones read-only. Each speaker listens on its own multicast group, one group per zone it belongs to, and the global all-call group. A Fanvil has 20 paging slots, so a speaker can be in up to 18 zones.

Automatic device reconfiguration

On supported devices (Fanvil) the paging list is rewritten in the background, only the paging entries, one request at a time per device, whenever a zone's members, address or name, a speaker's own group, the paging volume or the device's access details change. Other brands are configured by hand on the device (the Preview button shows the addresses to enter).

"Not in sync" badge

If a device doesn't accept its new list (offline, rebooting, wrong web credentials), the Speakers tab marks it **Not in sync** (the tooltip gives the reason) and admins get a login alert. The list is pushed again automatically as soon as the device answers the reachability check, which runs every 5 minutes.

Deleting a speaker empties its device

Deleting a reachable Fanvil speaker also empties the device's multicast list (its own group, its zones and the all-call group), so it stops playing announcements. If the device doesn't answer, or for other brands, a message says to reconfigure it by hand from its web page.

Protected deletions

Deleting a zone, speaker or audio file that a schedule uses is refused, with the list of schedules to fix first.

9.2 Live announcements

Conflict dialog

Starting an announcement on speakers that are already playing (another announcement or a schedule) asks first: **Stop them and play**, **Play alongside**, or cancel. The dialog also lists the schedules due while the announcement plays.

Bells wait for a live announcement

A live announcement always wins: a schedule due while it plays on some of the same speakers waits for it to end (up to 60 s) and then plays, instead of cutting in. If it would have to wait longer it is recorded as failed in the playback history (and counted in the login alert).

9.3 Schedules, holidays and custom dates

Per-country holidays, per schedule

Each schedule can use the holiday calendar of Italy, the United States, the United Kingdom, France, Germany, Spain, Japan, or China — exclude holidays, always include them, or play only on holidays. Holiday filtering is off by default.

Custom dates

In the Schedules tab: named lists of days or date ranges (closures, exam days, events), each entry optionally marked **Every year** (e.g. Dec 24 - Jan 6); dates can also be pasted one per line. A schedule can **Skip these days** or play **Only on these days**.

No overlapping schedules

A schedule lasts from its start time to the end of its audio file. Saving one that would play on the same speakers (directly, through a zone or via "all") at the same time as another enabled schedule is refused, naming the schedule in the way; the check looks two years ahead, day by day, with the same rules the scheduler uses. Overlaps that already existed (e.g. from before version 1.6.0) don't block unrelated edits; they, and overlaps created later by zone members or custom dates, are flagged in the list and reported in a login alert.

Timezone

Schedules fire in the timezone applied from System (or TIMEZONE from .env until one is chosen).

9.4 Interface

Light / dark / automatic theme

Selectable by anyone from the top bar. Automatic follows real sunrise/sunset at the coordinates of the configured timezone's reference city (Rome if none matches) — not the browser's geolocation, which is disabled by the dashboard's Permissions-Policy.

Multilingual interface

English (default), Italian, French, German, Spanish, Japanese, Simplified Chinese. Preference saved in the browser, not shared between users. Error messages from the server are shown in the viewer's language too.

No internet needed, strict Content Security Policy

Bootstrap and its icons are served by ZoneCast itself, so the dashboard works on an isolated LAN. Only scripts from the server itself may run (no inline scripts), which also blocks anything injected into the page.

Mobile

The whole dashboard works from a phone: sidebar drawer, adaptive tables and icon buttons.

Speaker brand dropdown

A list of the main multicast brands with an "Other" option for unlisted devices.

Audio analysis

Every uploaded file is analyzed for bass/mid/treble balance; a red badge warns if the bass content is high and might sound poor on a PA horn, and a gain amplification is suggested when there's headroom.

Host resource monitoring

Disk, RAM and CPU in real time in the System panel, with a color-coded bar.

Sessions and sign-in

A session expires after 12 hours without use (SESSION_MAX_AGE_SECONDS): a page left open doesn't keep it alive. After 5 failed attempts in 5 minutes, that username is locked out from that computer for up to 5 minutes. The default admin can't be deleted or made an Operator, so there is always an administrator.

Version and changelog

Clickable version number in System > Information, opens the change history.

9.5 Login alerts

After login, administrators see a banner when:

- scheduled playbacks failed in the last 24 hours;
- disk space is running low or is critically low;
- speakers didn't accept their latest zone setup (Not in sync);
- pairs of schedules play on the same speakers at the same time;
- the last automatic backup failed, no automatic backup has completed recently, or automatic backups are paused because their settings came from another installation (section 13).

Automatic backup (System tab, admin only) is described in section 13; the **network and time** panels (IP/DNS/gateway with a trial period, timezone, NTP servers, manual clock) are fully available on the native install only.

10. Migrating data from another installation

> **VERSIONS FIRST**
>
> Upgrade every installation (section 14) **before** moving bundles between them. A bundle exported by a newer ZoneCast version is refused at import ("This export comes from a newer ZoneCast version"); a bundle from an older version is fine — its database is upgraded automatically at the next start.

Step 1 — Export from the source installation

From the existing dashboard: System > Export full configuration, set a strong password and download the generated .zcbundle file. It contains the database, the encryption key (secret.key), media and device backups — treat it as sensitive material. An automatic backup file (section 13) has the same format and can be used instead, with the backup password.

Step 2, option A — Import from the dashboard

On the target installation, as an admin: System > Import configuration, choose the file, enter its password and press Import and restart. The bundle is decrypted and checked first (password, safe contents, schema version); only then is it queued, and the service restarts itself and applies it before opening the database. What it replaces is set aside next to it: `zonecast.db.pre-import-<time>`, `secret.key.pre-import-<time>` and, only if the bundle carries them, `media.pre-import-<time>` / `backups.pre-import-<time>`. A bundle without audio files leaves the existing ones in place.

Step 2, option B — Import from the command line, with the service stopped

Useful on a new machine, before the service's first real use. The bundle file must be readable by the zonecast user (native) — delete it afterwards. The tool asks for the export password. On the native install it runs as zonecast from inside /opt/zonecast, a folder closed to other users, hence the `sh -c`:

```
# --- Native install ---
sudo systemctl stop zonecast
sudo -u zonecast sh -c 'cd /opt/zonecast && \
  venv/bin/python -m app.tools.import_bundle /path/to/export.zcbundle'
sudo systemctl start zonecast

# --- Docker install ---
cd ~/zonecast && docker compose stop zonecast
docker run --rm -it \
  -v $(pwd)/data:/app/data -v $(pwd)/media:/app/media \
  -v $(pwd)/backups:/app/backups \
  -v /path/to/export.zcbundle:/tmp/export.zcbundle \
  zonecast-zonecast:latest \
  python -m app.tools.import_bundle /tmp/export.zcbundle --yes
docker compose up -d
```

> **IMPORTANT**
>
> The command-line import must be done with the service **stopped**: overwriting the SQLite database while the app has open connections risks corrupting it. Like the dashboard import, it refuses a bundle from a newer ZoneCast and first copies the current data aside (`*.pre-import-*`).

Step 3 — After the import

- The encryption key comes with the bundle, so device passwords, 2FA and saved backup passwords keep working.

- An automatic-backup policy that arrives with a bundle from another machine is **paused** (data/instance.id differs and is not part of the bundle), with a login alert: check it in System > Automatic backup and save it again. This avoids two machines backing up under the same settings by accident.
- Check the Speakers tab: devices that were not reachable from the new machine show Not in sync until they answer.

11. Security: 2FA, firewall, privileged helper, systemd hardening, HTTPS

Two-factor authentication (2FA)

Any user can enable it from the Security button: QR code for an authenticator app, 6-digit code, 10 recovery codes shown only once.

Firewall (ufw)

```
sudo ufw allow OpenSSH
sudo ufw allow 8000/tcp   # or whatever port APP_PORT is set to
sudo ufw enable
```

ufw allows outgoing traffic by default. If outgoing traffic is filtered (on the host or by a network firewall), ZoneCast needs: multicast UDP to the 239.x.x.x groups, HTTP to the speakers' web interface (provisioning, backups, reachability), NTP, and for automatic backup copies TCP 21 plus the FTP server's passive port range (FTPS/FTP — passive mode is always used) or TCP 445 (SMB).

How privilege escalation works for Network/Time

The application process always runs as the unprivileged `zonecast` user. Actions that need root privileges (changing the IP, timezone, NTP servers) go through a single helper, `/usr/local/sbin/zonecast-netctl` (installed from deploy/zonecast-netctl.sh, owned by root, mode 750), invoked via `sudo -n` and authorized by the rule in `/etc/sudoers.d/zonecast-netctl`, **scoped to that one path** (never arbitrary commands). The installer validates the rule with `visudo -c` before installing it, since a malformed file in /etc/sudoers.d would break sudo system-wide.

Since version 1.5.8 the helper lives outside the application folder and **the code under /opt/zonecast is owned by root**: the zonecast user can write only data/, media/ and backups/. If it could modify the code or the helper, any flaw in the app would become root access. Installations from before 1.5.8 get this layout by re-running the installer (section 14); until then the app keeps using the old helper path.

> **TECHNICAL NOTE — systemd hardening and sudo**
>
> The `deploy/zonecast.service` file deliberately does **not** set `NoNewPrivileges`, `ProtectSystem=strict` or a narrowed `CapabilityBoundingSet`, contrary to the more common hardening advice: all three would prevent sudo from escalating privileges inside it (the first blocks setuid escalation outright; the second would make /etc read-only even for the root child process sudo creates, since mount namespaces are inherited; the third caps the escalated child too, and sudo-rs — Ubuntu 26.04's default — needs CAP_SETUID). What keeps the app from writing outside its data is plain Unix ownership (see above); `ReadWritePaths` in the unit documents that set but doesn't enforce it without ProtectSystem. The unit does set ProtectHome and PrivateTmp, and grants only CAP_SYS_TIME and CAP_NET_BIND_SERVICE as ambient capabilities.

Dashboard and data

- Every response carries a strict Content Security Policy (scripts only from the server itself), X-Frame-Options DENY, X-Content-Type-Options nosniff and a same-origin Referrer-Policy; names, file names and log messages are always rendered as text.
- Passwords are hashed with bcrypt; device web passwords, 2FA secrets and automatic-backup passwords are encrypted with data/secret.key. Export bundles and automatic backups are encrypted with their own password.
- Python dependencies are installed from the hash-locked requirements.txt.
- Automatic backups: the backup password must be at least 12 characters; FTPS certificates are verified (or pinned by fingerprint); plain FTP requires explicit consent; SMB transfers are encrypted by default and DFS referrals are ignored, so credentials are never sent to another host (section 13).

HTTPS (optional)

The app speaks plain HTTP. To add TLS, put nginx in front of the app using the provided `deploy/nginx-zonecast.conf` template (it proxies to 127.0.0.1:8000), then certbot for a Let's Encrypt certificate (with a domain) or a self-signed certificate (LAN IP only). Set HSTS at the proxy if you want it: the app doesn't, since it can't know whether it's behind TLS.

12. Day-to-day operations and useful commands

Action	Native (systemd)	Docker
Status	systemctl status zonecast	docker compose ps
Live log	journalctl -u zonecast -f	docker compose logs -f zonecast
Restart	systemctl restart zonecast	docker compose restart zonecast
Stop	systemctl stop zonecast	docker compose down
Upgrade (section 14)	sudo ./deploy/install_ubuntu.sh from the new release	docker compose up -d --build in the updated folder
Pre-upgrade DB copies	sudo sh -c 'ls -l /opt/zonecast/data/zonecast.db.pre-upgrade-*'	ls -l data/zonecast.db.pre-upgrade-*
Automatic backups on this server	sudo ls -l /opt/zonecast/data/auto_backups/	sudo ls -l data/auto_backups/

Application logs (logins, provisioning errors, playbacks, backups) are always available and exportable from the dashboard: Log section (admins only). Automatic backups can also be started, listed, downloaded and deleted from System > Automatic backup.

13. Backup: manual export and automatic backup

13.1 Manual backups

- **Individual device config backup**: from each speaker's card, Backup button (admins)
- **Full application data backup**: System > Export full configuration — database, encryption key, media and device backups in one encrypted file; keep it off the machine
- **Database-only backup** (consistent even while the service runs, WAL included): `sudo -u zonecast sqlite3 /opt/zonecast/data/zonecast.db ".backup /opt/zonecast/data/zonecast.db.bak-$(date +%Y%m%d)"` — keep data/secret.key with it, or the encrypted values in it can't be read

13.2 Automatic backup (System tab, admin only)

The same encrypted bundle as Export full configuration, made on a schedule and kept on the server, optionally also copied elsewhere. Settings:

Setting	Meaning
How often	Every day or every week (on a chosen day), at a chosen time, in the schedules' timezone. A run missed while the service was down is caught up two minutes after the next start
Include the audio files	Leave them out to keep backups small (and under BACKUP_MAX_BUNDLE_MB); restoring such a backup leaves the target's audio files in place
Copies kept on this server	0-10 (default 3), in data/auto_backups/. At least 1 when there is no other destination; 0 keeps only the remote copy
Copies kept at the destination	1-100 (default 14)

Setting	Meaning
Backup password	At least 12 characters, and you must confirm you have stored it somewhere safe: **without it no backup can be restored**. Stored encrypted with data/secret.key, never shown again
Also copy to	Nowhere else, FTPS, plain FTP or an SMB share — see 13.3

How a run behaves, since it runs unattended on a PA controller:

- The bundle is built in a separate, low-priority child process, never in the process that paces the audio.
- A scheduled run is postponed (up to 30 minutes) while audio is playing or a schedule is due within two minutes.
- It is refused if its estimated size exceeds BACKUP_MAX_BUNDLE_MB (default 500) or if free disk space is less than twice that size plus 512 MB.
- Files are named `zonecast-<installation id>-<UTC time>Z.zcbundle`. Uploads go to a hidden temporary name, are renamed when complete and their size is checked.
- Retention runs only after a successful upload and **only ever deletes this installation's own files** (by the id from data/instance.id), so several installations can share one destination; it never deletes the file just written.
- Buttons: **Test connection** (connects, writes and deletes a small test file), **Back up now**, and the list of backups on this server with download and delete. Status, last successful backup, next run and the last error are shown in the same card.
- Login alerts (admins) when the last backup failed, when none has completed for about two periods, or when the policy is paused because it came from another installation.

13.3 Destinations

Destination	Details
This server only	data/auto_backups/ — protects against mistakes, not against losing the machine
FTPS (FTP with encryption)	Explicit TLS (AUTH TLS), port 21 by default. The certificate must be trusted by the system and match the server name, or — for a self-signed certificate — its SHA-256 fingerprint is pinned: Test connection shows the fingerprint of an untrusted certificate and **Trust this certificate** pins it, after you compare it with the one shown on the server. Passive mode, with the TLS session reused on the data connection (as vsftpd and FileZilla Server require). Firewall: outbound 21 plus the server's passive port range
FTP without encryption	Only after ticking the explicit consent: the login password and the file travel unencrypted (the file itself is still protected by the backup password). Use FTPS whenever possible
SMB share (SMB2/3)	Port 445, NTLM authentication; the username can be `user` or `DOMAIN\user`. **Encrypt the transfer (SMB 3)** is on by default — untick it only for a server without SMB 3 encryption. DFS referrals are ignored (credentials never follow a referral to another host): enter the real server and share, not a DFS namespace. The folder is created if missing

Server, share and folder names accept letters, digits, dots, dashes, underscores and spaces (no paths such as \\server\share). When the server, port, share or username change, the password must be typed again: a saved password is never sent to a different server.

13.4 Restoring and moving

- Restore = **System > Import configuration** with the file (downloaded from the list, or taken from the destination) and the **backup password**; or the command-line import (section 10).
- On the same installation the policy stays active; on another machine it arrives paused and must be saved again (data/instance.id).
- If data/secret.key changes — e.g. the database was moved without its key, or the key was replaced — the saved backup passwords can't be read any more: runs fail with "The saved backup passwords can't be read with this installation's key". Re-enter the backup password and the destination password in System > Automatic backup and save. Existing backup files still open with the password they were made with.

14. Upgrading and rolling back

14.1 Before upgrading

- Read the changelog of the new release (System > Information shows the installed one).
- Take a manual export (System > Export full configuration) and keep it off the machine.
- Pick a moment without bells: the service restarts during the upgrade.
- Upgrade every installation before moving bundles between them (section 10).

14.2 Native install

Get the new release (install kit or repository) into a fresh folder and re-run the installer from there:

```
mkdir -p /tmp/zonecast-new
tar -xzf zonecast_install_kit.tar.gz -C /tmp/zonecast-new
cd /tmp/zonecast-new
sudo ./deploy/install_ubuntu.sh
```

On an existing installation, a re-run:

- replaces app/, migrations/, deploy/, alembic.ini and requirements.txt in /opt/zonecast with the new ones (rsync --delete, so files removed from the release disappear too);
- keeps .env (it is only created when missing) and never touches data/, media/ or backups/;
- reuses the venv and installs the new hash-locked requirements into it, then recompiles the bytecode;
- reapplies ownership (code root-owned, only data/, media/, backups/ writable by zonecast), reinstalls /usr/local/sbin/zonecast-netctl and its validated sudoers rule, and the systemd unit;
- restarts the service: at start the schema migrations run automatically (14.4).

14.3 Docker install

Replace the project files in ~/zonecast with the new release, keeping .env, data/, media/ and backups/, then rebuild and restart:

```
cd ~/zonecast
docker compose up -d --build
```

14.4 What happens at start: schema migrations

Database schema migrations (Alembic) run **automatically at every start**, in a single transaction, before anything else opens the database — on the native install, with Docker, and after importing a bundle from an older version. Installations from before Alembic (up to 1.5.7) are recognised and upgraded too. Before an upgrade the database is copied next to itself as `data/zonecast.db.pre-upgrade-<revision>-<time>` (revision "legacy" for a pre-Alembic database); the newest 3 copies are kept, and older ones are removed only after an upgrade succeeded. A start with nothing to upgrade makes no copy.

14.5 Rolling back

The older release can't use a database upgraded by the newer one (it refuses to start). Restore the pre-upgrade copy, then reinstall the previous release. Changes made after the upgrade are lost.

```
# --- Native install ---
sudo systemctl stop zonecast
sudo -s                 # root shell: /opt/zonecast is closed to other users
cd /opt/zonecast/data
ls -lt zonecast.db.pre-upgrade-*          # pick the copy to restore
# set the upgraded database aside (with its -wal/-shm, if present)
for s in "" -wal -shm; do
  [ -e zonecast.db$s ] && mv zonecast.db$s zonecast.db.failed-upgrade$s
done
cp -p zonecast.db.pre-upgrade-<revision>-<time> zonecast.db
exit
# then, from the folder of the PREVIOUS release:
sudo ./deploy/install_ubuntu.sh           # reinstalls it and starts the service

# --- Docker install ---
cd ~/zonecast && docker compose down
# same file operations as above, in ./data
# put back the previous release's files, keeping .env and the data folders
docker compose up -d --build
```

14.6 Upgrading from a version before 1.5.8

Re-running the installer also moves the privileged helper to /usr/local/sbin/zonecast-netctl, rewrites the sudoers rule for it and makes the code root-owned (section 11). Until it is re-run, the app keeps using the old helper at /opt/zonecast/deploy/zonecast-netctl.sh.

15. Common troubleshooting

Symptom	Possible cause / fix
Audio doesn't reach the speakers (Docker)	Check network_mode: host in docker-compose.yml and that the switch isn't filtering multicast/IGMP traffic
A speaker shows "Not in sync"	The device didn't accept its latest zone setup (offline, rebooting, wrong web credentials); the tooltip gives the reason. It is pushed again automatically when the device answers the next reachability check (every 5 minutes). Fix the credentials in the speaker form if needed, or force it with Apply now (admin); Preview shows what should be on the device
A speaker can't be added to one more zone	"... is already in 18 zones": a Fanvil has 20 paging slots — its own group, the all-call group and up to 18 zones
Saving a schedule is refused because it overlaps	"It would play on the same speakers at the same time as ... other schedule(s)": a schedule lasts as long as its audio file. Change the time, days, date range, custom dates or destination of one of the two, or disable the other one
Schedules flagged as overlapping / login alert about pairs of schedules	Overlaps that existed before 1.6.0, or were created later by zone members or custom dates. Edit one schedule of each pair
A scheduled playback failed during a live announcement	It waited the maximum 60 s for the announcement on the same speakers to end, then was recorded as failed. Keep live announcements shorter around bell times
A deleted speaker keeps playing announcements	When it was deleted the device wasn't reachable (or isn't a Fanvil), so its multicast list wasn't emptied: remove the multicast groups on its web page or reset it to factory settings
Deleting a zone, speaker or audio file is refused	Schedules use it: the message lists them. Edit or delete them first
"The device did not return a login nonce"	The Fanvil's web server is temporarily overloaded or rate-limiting — wait a few minutes and retry from Apply now. Don't send rapid repeated requests to the same device
A changed IP stays reachable at the old address too	Check that no other file in /etc/netplan/ declares the same address besides 90-zonecast.yaml — ZoneCast disables pre-existing files on first apply, but an installation with multiple network interfaces may need manual review
NTP server set but status stays "Not synced"	Normal for the first 30-90 seconds after changing servers: chrony needs a few polling cycles before declaring itself synced. Also check the hostname is correct (e.g. time.google.com, not ntp.google.com — the latter doesn't exist)
Network/Time panels are read-only or fail	Read-only on Docker by design. On a native install, re-run the installer so /usr/local/sbin/zonecast-netctl and its sudoers rule are in place
An operator can't use Apply now/Backup or see Users, System, Log, Backups	By design: those (including automatic backup and the login alerts) are admin-only. Operators can manage zones, speakers, media, schedules and custom dates
Backup test/run: "The server refused the username or password"	Check the credentials; for SMB try `DOMAIN\user`. After changing server or username the password must be typed again
Backup test: "The encrypted (TLS) connection failed"	The FTPS certificate isn't trusted or doesn't match the name: compare the fingerprint shown with the server's and use Trust this certificate, or install a certificate from a trusted CA. The server must support explicit FTPS (AUTH TLS) on that port

Symptom	Possible cause / fix
"The server's certificate is not the one that was trusted"	The certificate changed (renewed) — or the connection is being intercepted. Verify on the server, then pin the new fingerprint
"The share or the folder doesn't exist"	Enter only the share name (not \\server\share) and the real server: DFS namespaces are not followed
"Couldn't connect to the server" / upload fails after login	Outbound TCP 21 plus the passive port range (FTP/FTPS) or TCP 445 (SMB) blocked by a firewall; for an SMB server without SMB 3 encryption untick Encrypt the transfer
"Not enough free disk space on this server for the backup"	A run needs twice its estimated size plus 512 MB free: keep fewer local copies, leave out the audio files, or free disk space
"The backup would be too big"	Leave out the audio files, or raise BACKUP_MAX_BUNDLE_MB in .env and restart the service
"The saved backup passwords can't be read with this installation's key"	data/secret.key is not the one the passwords were saved with (database moved to a machine with another key, or key replaced). Re-enter the backup and destination passwords in System > Automatic backup and save
Automatic backups paused: "their settings came from another installation"	The policy arrived with an imported bundle. Check it in System > Automatic backup and save it again
Docker: the container stops at once with "... is not writable by the user ZoneCast runs as"	The data, media or backups folder belongs to root (Docker created it because it was missing): next to docker-compose.yml run sudo chown -R 1000:1000 data media backups, then docker compose up -d
Import refused: "This export comes from a newer ZoneCast version"	Upgrade this installation first (section 14), then import again
The service doesn't start after a rollback	The log says the database is at a revision newer than this version: install the newer release again, or restore the zonecast.db.pre-upgrade-* copy (section 14.5)
Error importing a bundle	"Wrong password, or corrupted file": use the password chosen at export time (the backup password for automatic backups); for the command-line import the service must be stopped
After a user was deleted, their open browser kept bouncing between login and dashboard	No longer happens: the stale session is dropped and the login page is shown
"ModuleNotFoundError" after a major Ubuntu upgrade	Recreate the venv and re-run the installer (note in section 5)
Sent back to the login page without logging out	The session expired after 12 hours without use (a page left open doesn't count): sign in again. Change the length with SESSION_MAX_AGE_SECONDS in .env
"Too many failed attempts. Try again in N seconds"	5 failed sign-ins in 5 minutes with that username from that computer: wait the time shown (at most 5 minutes) or sign in from another computer
Admin password lost	On the server: sudo -u zonecast sh -c 'cd /opt/zonecast && venv/bin/python -m app.tools.reset_admin_password admin NewPassword' (creates the user if missing). That user's open sessions end
The default admin's role can't be changed	By design (since 1.6.1): it always stays an administrator, so there is always an account that can manage users and the system

Symptom	Possible cause / fix
Locked out after losing the 2FA authenticator app	Use one of the 10 recovery codes saved at activation; if lost, an administrator with server access can disable a user's 2FA in the database (users table: totp_enabled, totp_secret, totp_recovery_codes)

16. Appendix — list of files provided

File	Purpose
zonecast_install_kit.tar.gz	Application package for a native or Docker install — extract it on the new machine. Contains app/, migrations/, alembic.ini, deploy/, requirements.txt, Dockerfile, docker-compose.yml, .env.example, README.md
app/	The application (backend, dashboard templates, static files and vendored Bootstrap)
migrations/, alembic.ini	Alembic database schema migrations, applied automatically at every start
deploy/install_ubuntu.sh	Native install and upgrade script (Method A)
deploy/zonecast.service	systemd unit file for the native run
deploy/zonecast-netctl.sh	Source of the privileged helper for NTP/timezone/network; installed as /usr/local/sbin/zonecast-netctl and invoked via a scoped sudo rule
deploy/nginx-zonecast.conf	nginx reverse-proxy template for adding HTTPS
Dockerfile / docker-compose.yml	Image and orchestration for Method B
.env.example	Configuration file template
requirements.txt	Runtime dependencies, every package pinned with its hash (generated)
requirements.in	Direct runtime dependencies, from which requirements.txt is generated with uv
requirements-dev.in / requirements-dev.txt	Test and lint tools, same scheme
pyproject.toml	Tool configuration (ruff, pytest, coverage) — ZoneCast isn't installed as a package
app/tools/import_bundle.py	Script to restore a full export from the command line
app/tools/reset_admin_password.py	Resets (or creates) an admin's password directly in the database
README.md	Project's technical reference documentation
CONTRIBUTING.md	Development setup, dependency and schema-change workflow, tests